
Pump Back Hydro Feasibility Report

HENDERSON CHALLENGE

STORING ENERGY FOR A RENEWABLE FUTURE

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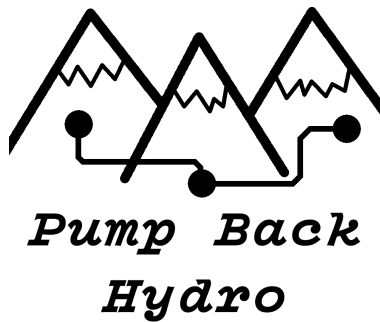
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RICHARD PALIN

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MEET THE TEAM:

The team members have a variety of experiences that are directly applicable to this proposed development. The team also has prior entrepreneurial experience and has successfully worked on previous projects and hackathons similar to this one.

Gage Coprivnicar is an aspiring Electrical Engineer with a focus in renewable energy. This applies to the project when running the calculations and estimations necessary for the transport, generation, and storage of the electricity.

Martina Gilbert is working towards a degree in Mining Engineering and was crucial to fully understanding the project's undertaking and interpreting the diagrams provided by Henderson Mine. She allowed for an understanding of exactly where the water would be running in the mine as well as the infrastructure changes necessary for the project to work.

Julian Paik, a Biochemical Engineering major with prior industry experiences allowed for a full understanding of what sort of incentives there were for renewable energy generation. Including but not limited to relevant grants and methods of finding the required funding for the project to be completed.

Lastly, Samuel Zowam with an bachelors degree in Petroleum Engineering and progress towards a masters in Energy Economics, built from scratch the teams necessary economic models that allowed for an accurate estimation of what the buying and selling of electricity looked like on a large scale; he also estimated relevant returns and total revenue and profit by creating his own economic model.

Our mentors for this project, Mr. Glenn Krum, and Dr. Richard Palin provided invaluable feedback to our team and assisted in much of the project. They also provided a large variety of resources that were utilized, and they each brought their own perspectives on the feasibility and benefits of the proposed project.

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1 Overview

1.1 Proposal

Pump Back Hydro (PBH) aims to store electricity using reservoirs and existing infrastructure at Henderson Mine as a way to enable a larger dependency on renewable electricity into the future. By storing electricity generated at peak supply times during the day in the form of water at a high elevation, it is possible to generate energy during off peak times when electricity is more expensive by allowing the water to fall a significant vertical distance through the generators. This method of storage has been used for many years and has been proven to be effective and efficient. With the proposed project, it is estimated that approximately 1,600MW of electricity, or enough to power 3.25% of Colorado's households, can be produced daily.

1.2 Utilized Assets

This project is centered around the existing altitude variation provided by the Henderson Mine. The configuration of the mine shafts provides the opportunity to introduce a pumped hydroelectric storage facility utilizing two hydroelectric generators. The Reclaim Gallery will be repurposed into an underground containment area for the water once it falls through the generators in the shaft. Buildings on the surface will be repurposed to accommodate Pump Back Hydro personnel.

1.3 Detailed Description

Our project plans on utilizing the unique landscape that is present at Henderson Mines. On the property there are a few key pieces that we will be using. On the Mine Site we plan on using shafts 1, 2, 3, and 5. The No. 1 shaft will be used as a way to transport the water up and down between the Reclaim Gallery and the Water Tank that will be built on the surface, see **Appendix**. The water tank will be a cylinder capable of holding $600,000\text{m}^3$ and be 20 meters in height with a radius of 97.7 meters.

Shafts No. 5 and 3 will be used as a way to ventilate the mine to allow the workers to maintain and check out the used areas. Much of the mine will be blocked off allowing for less expensive maintenance. Lastly the No. 2 shaft will continue to be used to transport workers and materials into the mine.

The generators will be placed at depths dependant on the existing infrastructure as a way to support the generator and simplify the installation process. The ideal are around 2330m and 2112m as installation would be the simplest and most cost efficient.

2 Market Overview

2.1 Target Customer

All of Pump Back Hydro's customers are on the Western Interconnection Grid and are a combination of businesses, individuals, and government facilities; this creates a diverse electricity demand across the grid. With this in mind, PBH does not sell directly end-users, but instead goes through grid operators that handle individual sales throughout the day.

2.2 Market Size and Growth

Nearly 75% of the U.S. has mandatory or voluntary expanding renewable energy goals, including Colorado. These goals present a challenge as most renewable energy is available during the day but a large portion of electricity is used at night. To reach renewable energy goals, it is imperative to store excess electricity generated during the day for nightly use.

With these renewable goals in mind, Henderson Mine is connected to the larger Western Interconnection Grid which consists of eleven full states and portions of Canada and Texas. Many of these states have renewable energy goals as demonstrated in **Figure 1**. This allows for a large market to buy and sell electricity from with an opportunity to make profit and with growth in the renewable energy sector Pump Back Hydro energy storage is viable well into the future.

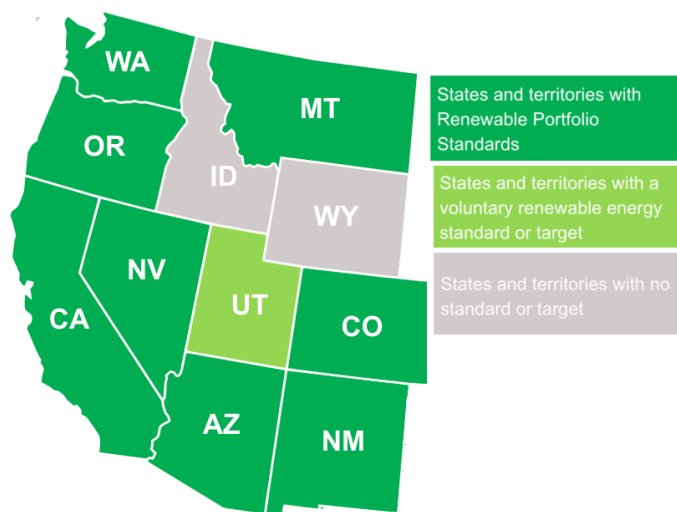


Figure 1: Western Interconnection Renewable Portfolio Standards, *Provided by NCSL*

To be a viable business, continuing operations would need a market share of roughly 1.2% based on per-capita electricity consumption and the Western Interconnection servicing a population of 80+ million. This, tied to the United State's forecasted 1.1% increase in electricity consumption annually, supports long term viability. It is also estimated that the facility will be able to sell electricity at a price difference of \$0.1368. This all factors into the BTCF and other details which are shown below.

In addition to the aforementioned, pumped hydroelectricity is also the fastest form of electricity production; unlike other fossil fuels like coal and petroleum, pumped hydroelectricity can produce electricity near immediately meaning it can fulfill spikes in consumer demand quicker than any other energy source. This allows the facility to respond to blackouts and brownouts in surrounding areas quicker further increasing the viability of the plant.

3 Business Model

When analyzing the business's viability and figures as shown in **Table 1**, it is important to understand a few of the basic assumptions used to build this model. It is estimated that facility is capable of generating 1,393MW a day and that the facility has the necessary 1.2% market share across the Western Interconnection. Pump Back Hydro also estimates that it will need to employ 70 employees with an average pay of \$77,500 annually.

Table 1: PBH Business Model

Year	0	5
Electrical Price Diff	\$0.1368	\$0.1396
Electrical Production/Year	508,695,666	508,695,666
Gross Revenue	\$69,589,567	\$70,992,537
Operation Costs	\$(19,405,000)	\$(19,502,219)
BTCF	\$(436,815,433)	\$51,490,318
Taxes Generated	\$0	\$3,286,954
ATCF	\$(436,815,433)	\$48,203,364

4 Development Details

4.1 Projected Capital Costs

The projected capital costs to build the described project comes to \$487,000,000. The breakdown of this into categories is shown in the table below. Prices are from other pumped hydroelectric facility projects and quotes received from General Electric for industrial water power equipment. This estimate does not include any project incentives; these are detailed in the following section.

Table 2: Pumped Hydroelectric Facility Development Costs

Description	Amount	Percent
Reservoir	\$189,930,000	39%
Powerhouse, Generator	\$48,700,000	10%
Grid Connection, Converters	\$4,870,000	1%
Engineering, Construction Management	\$102,270,000	21%
Owner's cost	\$73,050,000	15%
Tunnel/HVAC	\$68,180,000	14%
Total	\$487,000,000	100%

4.2 Funding Opportunities

Please note that the below economic incentives will vary depending on political climate and demand. Given historical trends and future hydropower growth, we are making the assumption that these federal and local incentives will remain the same.

4.2.1 Federal Incentives

The U.S. Department of Energy has several federal funding opportunities for water power projects. Under section 242 of the Energy Policy Act of 2005, hydroelectric facilities can receive up to ¢1.8 per kWh of electricity produced. This subsidy is limited to \$750,000 annually for a maximum period of 10 years and would be fully utilized based on anticipated power production. The Henderson facility would also be eligible for the Renewable Electricity Production Tax Credit (PTC) or the Business Energy Investment Tax Credit (ITC). The PTC is tied to the facility's annual electricity production, equal to ¢1.1 per kWh over a 10 year period, and the ITC is equal to 30% of

the facility's value. Based on expected electricity production and start-up costs, the ITC provides a greater tax credit. With these incentives, an estimated \$153,000,000 will contribute to the facility ROI and payback period. This essentially reduces the initial investment by 31%.

4.2.2 Colorado Incentives

The Colorado Energy Office (CEO) also has different incentives for associated projects. While Colorado does not offer local tax credits or subsidies, the CEO supports renewable energy production through expedited project permitting and consumption mandates, more commonly known as the Renewable Energy Standard (RES). Expedited permitting helps to get renewable energy projects started quicker than traditional electricity production projects saving in facility downtime. The CEO RES also mandates that investor-owned utilities produce a certain percentage of renewable energy as part of their total annual energy production. This impacts project funding by creating a market that entices private investments and loans.

4.2.3 Other Initiatives

In addition to federal and local assistance, there are multiple national organizations, such as the National Hydropower Association and the Colorado Small Hydro Association that provide training programs for individuals pursuing employment in the hydroelectric industry.

4.3 Development Schedule

In general, for a large scale pumped hydroelectric facility (above 1200MW capacity), there is a permitting timeline of 1-3 years followed by construction typically completed within 3 years. Based on other pumped hydroelectric facilities of this size and in geographic regions similar to Henderson, assume a 4 year completion timeline outlined in **Figure 2** below.



Figure 2: Colorado Hydropower Permitting and Construction Process, *Provided by RAPID*

Initial project permitting can begin far before Henderson Mine’s closure. The construction of the above ground reservoir could begin as soon as the area of operation is vacant and could even be constructed off-site, then moved. Construction of the lower reservoir would begin at this time as well and would be followed by installing the generators and the pipes and routing them between the upper and lower reservoirs.

5 Impact and Risks

5.1 Surrounding Impact

5.1.1 Direct Impact

There are several direct impacts that this new facility brings to the community. This includes taxes paid to the community, employee salaries, and electricity subsidizes. In total it is estimated that this comes to \$25 million.

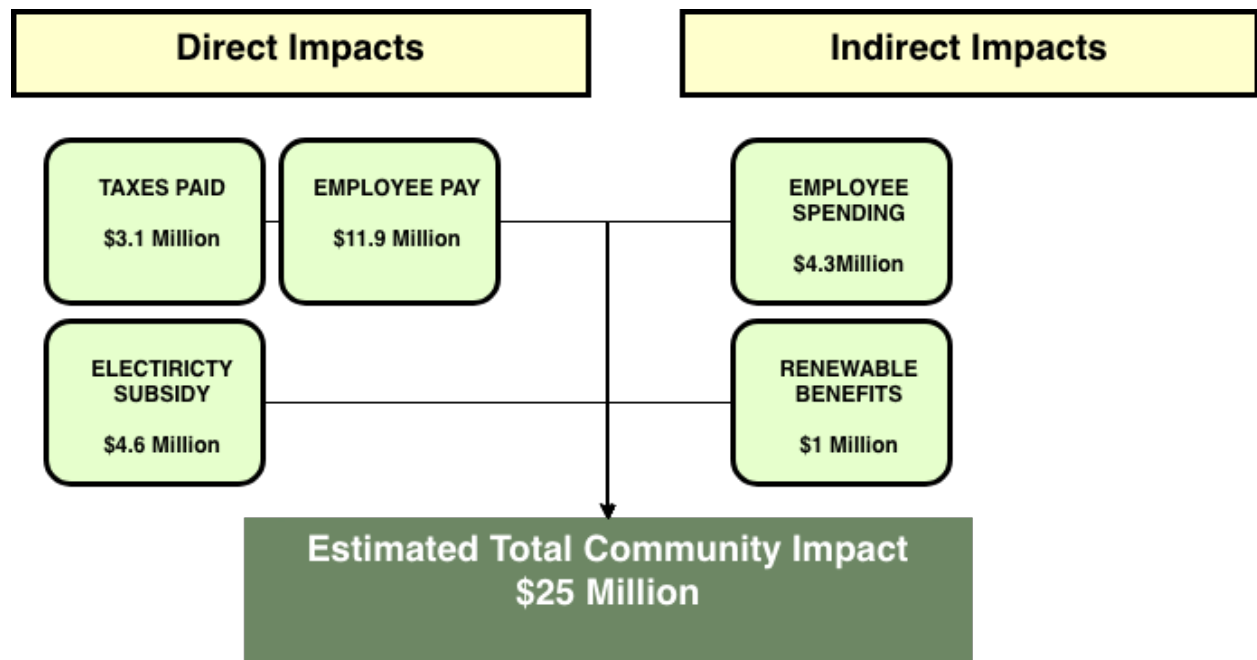


Figure 3: Economic Impact on Surround Communities

5.1.2 Indirect Impact

Instead of using Henderson’s native water sources of the URAD Reservoirs, PBH opted for a closed loop system to minimize negative environmental impacts in the surrounding area. A closed loop system has specific benefits such as: resistance to climate changes/ambient temperatures, separate from biological factors such as wildlife and various enzymes, more accurate monitoring of water pressure and volumes/flow rates.

5.2 Risks and Mitigation

The risks of establishing an energy storage plant, condensed into a risk mitigation table shown below.

Table 3: Associated Risks of Operation

Risk	Mitigation	Likelihood /Severity (1-5)
Water Tank Burst	Keep staff ready to repair and enable emergency shutoff of the water supply. Schedule regular maintenance.	2/3
Generator Failure	To prevent generator failure it will be regularly maintained and debris will be cleared out of the water to prevent blockages.	1/4
Electricity Demand Falls	This is a difficult situation to cope with but with the general trends in electrical demand it is unexpected. If it does happen the company has to accept lesser profits.	2/2
Underground Pipe Burst	Regular maintenance will be utilized to ensure that an underground pipe will not burst	2/3
Underground Disaster	Pumped Back Hydro will comply with all state and federal regulations regarding underground work, health, safety, and emergency response.	1/5

6 Citations

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7 Appendix - Facility Design, Technical Drawings

